Predicting English Language Recall in Spaced Repetition Learning using Logistic Regression and Random Forest

Abstract**—**This project report will present a prediction model on student recall of English language vocabulary based on historical interaction metrics and temporal spacing. Measurements were taken from Duolingo's spaced-repetition dataset, tracking individual practice sessions [6]. The two machine learning methodologies used are the Logistic Regression model and the Random Forest algorithm. By utilizing these methods this project will attempt to estimate the probability that a learner will successfully recall a word given their historical performance and the elapsed time since their last interaction.

Keywords—Logistic Regression, Random Forest, Spaced Repetition, Memory Decay

I. INTRODUCTION

The optimization of vocabulary retention is a core challenge in second language learning. Online platforms such as Duolingo utilize spaced repetition systems to schedule practice sessions, operating on the principle that memory decay can be mitigated by reviewing information at strategically spaced intervals [1]. While specialized algorithms have been developed to model this memory decay, there is value in assessing whether standard, generalized machine learning algorithms are able to achieve comparable predictive utility using raw interaction data.

This project investigates whether we can accurately predict student word recall; specifically whether a language learner will remember or forget a specific word during a practice session based on the number of past practice sessions and the time that has elapsed between them. The primary aim of this report is to evaluate and contrast the predictive performance of two supervised learning models on a subset of Duolingo's spaced-repetition dataset [6]. By framing this as a binary classification problem, the project aims to determine which model best handles the complexities of the language learning data set, including disparities in feature scaling and non-linear relationships.

This report hypothesizes that a student's historical success rate with a word and the time elapsed since their last review (the spacing effect) will serve as the strongest predictors of successful recall, based on the premise of the human forgetting curve, which illustrates the way in which human retention of words declines over time, with the sharpest decline occurring within the first 24 hours after a word is learnt, and further decline in the following days resulting in a loss of roughly 50-90% of new knowledge learnt [8]. By applying these foundational algorithms and established psychological concepts, this report will assess the predictive power of spaced-repetition metrics, and determine whether standard classification techniques can effectively model word retention without relying on platform-specific algorithms.

OBJECTIVES

- Determine which features are most strongly associated with successful word recall
- Utilise two machine learning methods to produce an accurate predictive model for word recall based on relevant features
- Evaluate and contrast the accuracy of given classification models, determining their strengths and weaknesses

II. LITERATURE REVIEW

The application of machine learning to optimize educational scheduling is deeply rooted in cognitive psychology, specifically concerning the spacing effect. Research demonstrates that spacing out repeated encounters with learning material produces superior long-term retention compared to mass practice [7]. This phenomenon is particularly critical in second language acquisition, where reviewing vocabulary at strategically spaced intervals significantly boosts memory consolidation. In this section, the problem will be placed in the context of related studies in existing literature.

Firstly, the foundational study for this dataset, 'A Trainable Spaced Repetition Model for Language Learning' by Settles and Meeder [1], examined the impact of the spacing effect in relation to the primary factors that determine the probability of word recall. To address this problem they proposed the introduction of a machine learning algorithm called Half-Life Regression (HLR), a tailored model designed to estimate memory decay, and estimate the half-life of a word in a student's memory by analyzing data on practice history features from the Duolingo platform. Whilst this specialized model achieved high accuracy for this specific software architecture, the researchers noted that it required a complex, custom-built architecture specifically tuned to the physics of memory [1]. This presents an opportunity to investigate whether more generalized machine learning algorithms could achieve comparable predictive utility on the same dataset without such specialized modeling.

The second study, titled ‘Enhancing human learning via spaced repetition optimization’ by Behzad Tabibian et al. [2], investigated human memory models to gain a better understanding of how to optimize review schedules from a data-driven perspective. Their research specifically aimed to analyze the human forgetting curve, and model the exact probability that a learner could recall an item at any given time. Developing predictive algorithms based on learner's previous successful or unsuccessful recalls was discovered to be highly effective, in contrast to traditional spaced repetition models that typically relied on rigid, rule-based heuristics rather than personalized interaction data. By applying modern learning algorithms to the Duolingo dataset, this report aims to build upon the work of Tabibian et al. and test whether standard, non-linear models like Random Forest are better suited to capturing the nuances of human learning.

The third paper, ‘Personalized Language Learning Using Spaced Repetition Scheduling’ by Riku Fukui and Toshiyuki Ando [3], explores the spaced repetition technique and evaluates the effectiveness of tracking student memory states through discrete interaction logs. Their study highlighted a methodology in which learners provided binary feedback upon an attempt to recall a word’s meaning, measured as either a success or failure. This report will take a similar binary approach to test the predictive effectiveness of linear and non-linear models, and determine which better approximates the human forgetting curve to predict future recall.

The fourth study, ‘Predicting Students' Academic Performance Via Machine Learning Algorithms: An Empirical Review and Practical Application’ by Batool et al. [4], further justifies the transition from traditional statistical models to advanced machine learning techniques within the domain of educational prediction by demonstrating the Random Forest algorithms effectiveness in capturing the non-linear nuances of behavioral data. Their research provides a strong empirical basis for applying similar techniques to predict language recall, supporting the methodology of this report. Specifically, the findings of Batool et al. provide evidence that Random Forest is effective for classifying student performance because it can handle the complex, non-linear relationships often found in educational data, such as the relationship between practice history and memory recall, making the model better suited for the imbalanced nature of educational datasets.

The final study, titled ‘A logistics regression-based student performance prediction system,’ published in the Global Journal of Engineering and Technology Advances [5], provides a vital counterpoint by examining the link between behavioral indicators and binary classification outcomes. This research highlights that, while more complex learning models often prioritize raw accuracy, Logistic Regression offers an important balance between predictive performance and interpretability. Its inherent transparency allows for a clear understanding of the specific weights features have in influencing student success rate, justifying the use of Logistic Regression as a baseline to evaluate against the more complex, non-linear predictions of the Random Forest algorithm.

III. DATA MANAGEMENT

This project uses the Duolingo Spaced Repetition Dataset, which provides approximately 13 million rows of learning traces, with 12 different features [6], as shown in figure 1. Due to the large scale of the original dataset, a chunking strategy was implemented in order to process the data in blocks, with chunk size set to 500,000. The data initially contained results for learning multiple different languages. However, given the fact that some languages are inherently harder to learn than others, I decided to filter these out a focus solely on English learners, allowing my model to focus on the relationship between features and outcome. A random sample of 100,000 traces was then taken from this data, providing a statistically significant representation of learner behaviour whilst remaining computationally feasible to work with.

Variable	Description
p_recall	Proportion of exercises where word was correctly recalled
timestamp	UNIX timestamp of the current practice
delta	Time (in seconds) since the last practice that included this word
user_id	Student user ID (anonymized)
learning_language	Language being learned
ui_language	User interface language (presumably native to the student)
lexeme_id	System ID for lexeme tag (i.e. word)
lexeme_string	String representation of lexeme tag for the word/concept being tested
history_seen	Total times user has seen the word prior to this practice
history_correct	Total times user has been correct prior to this practice
session_seen	Times the user saw the word during this practice
session_correct	Times the user got the word correct during this practice

Figure 1: Dataset Features

Data exploration was first carried out in order to better understand the type of data present and determine which features were suitable for training a model. The initial analysis of the user_id column revealed a high number of unique users, with 23,577 unique individuals within the 100,000 row sample. Given this, I decided that it was best to drop the user_id tag and focus on general learning patterns across the sample population to avoid the risk of overfitting, modelling on broad patterns rather than focusing on specific individuals. I also decided to focus on historic session data over data from the current sessions in order to better account for the cumulative effects of spaced repetition, allowing my model to differentiate between transient short-term recall and established long-term knowledge.

Analysis of the target variable p_recall revealed that it was for the most part bimodal, with the majority of instances clustered around the extreme ends of the scale. As there was low data density in the middle ground, a regression model would struggle to find meaningful patterns if kept as a continuous variable, so I decided conversion to a binary binary classification task would allow my model to provide more actionable insight.

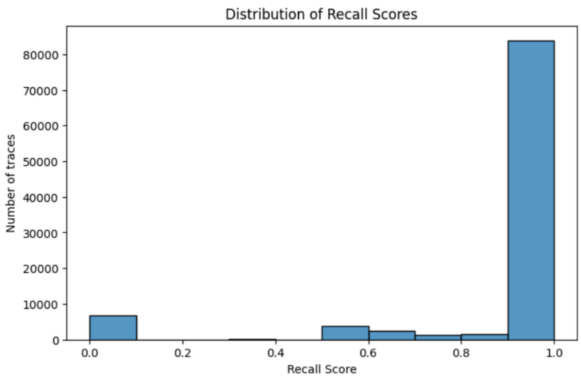


Figure 2: Distribution of p_recall

The distribution of my binary target variable, perfect_recall, showed a significant class imbalance, with approximately 83.7% of traces resulting in successful recall, and only 16.3% in mistakes. Given that apps like Duolingo are designed to keep users motivated by presenting achievable challenges, this high success bias was expected. However, this class imbalance suggested that my model would likely be biased towards predicting success and may struggle to identify instances of student failure, making it clear I would need to prioritize metrics that accounted for class imbalance over raw accuracy to ensure the model effectively identified the minority class.

```
Normalised counts for perfect recall:
perfect_recall
1    0.83686
0    0.16314
Name: proportion, dtype: float64
```

Figure 3: Distribution of perfect_recall

Univariate analysis of the delta feature revealed extreme right-skewness and a significant volume of outliers, with the data exhibiting a non-Gaussian distribution. This justified the application of a logarithmic transformation (delta_log) to normalize the feature scale. As illustrated in Figure 4, this converted the highly skewed raw distribution into a near-normal distribution, and revealed two main clusters indicating different modes of learner behavior, corresponding to immediate intra-session reviews and long-term spaced repetition intervals.

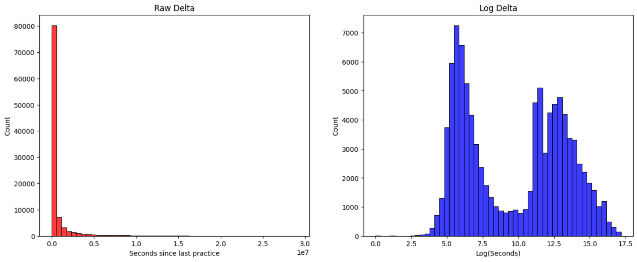


Figure 4: Distribution of raw delta vs Log delta

Bivariate analysis was used to visualize the relationship between learner's native languages and their success rate and determine whether ui_language, serving as a proxy to native language, should be prioritized as a predictive feature in the final model. The success-to-mistake ratio appeared relatively consistent across the language groups, suggesting that the underlying mechanics of the forgetting curve were relatively universal across learner's native language. Although this meant this was likely a weaker predictor of recall success than time-based or history-based features, I decided to retain ui_language in the model to act as a negative control, and converted the text categories into binary columns to ensure the dataset was in a suitable format for supervised learning.

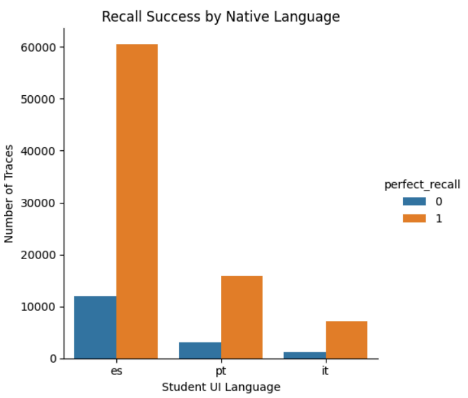


Figure 5: Recall success by native language

Initial bivariate analysis of history_seen against the target variable revealed a counter-intuitive trend, with the mean indicating that a higher number of prior sightings correlated with higher incorrect recalls. However upon closer inspection via boxplot analysis, it became evident that these averages were heavily skewed rather than representing a consistent trend. With the vast majority of data for both outcomes concentrated near zero, coupled with the extreme outliers, I decided to mitigate this issue by engineering the history_success_rate feature, calculating the ratio of successful past reviews to total sightings to compress the data into a bounded range.

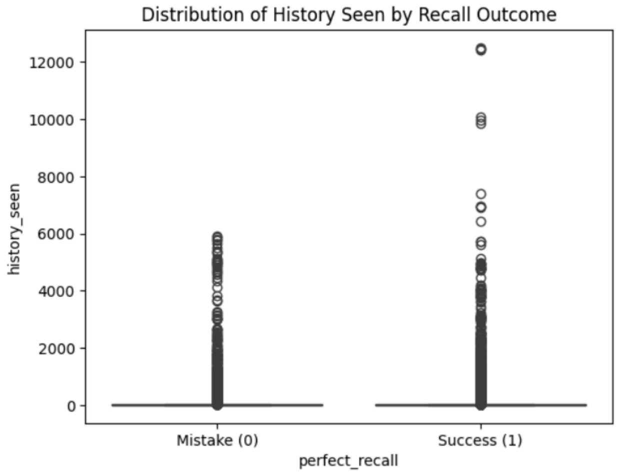


Figure 6: Recall success by average practice

Final feature selection was performed to ensure model stability and generalizability, with the remaining features dropped in favour of the engineered history_success_rate, the normalized delta_log and the categorical values for ui_language. To ensure a significance of the remaining features, four distinct importance metrics were employed: two correlations tests (Chi-Squared and F-test), and two machine learning techniques (Mutual Information and ExtraTrees), capturing both linear and non-linear dependencies and providing a comprehensive ranking of the predictors. Chi-Squared and ExtraTrees identified delta_log as the most effective feature, supporting the theoretical spacing effect, whilst the F-test and Mutual Information prioritized historical accuracy as the strongest indicator. Conversely, the consistent low ranking of linguistic background across all tests validated its role as a negative control, suggesting that cognitive behavioral patterns are far more predictive than demographic variables.

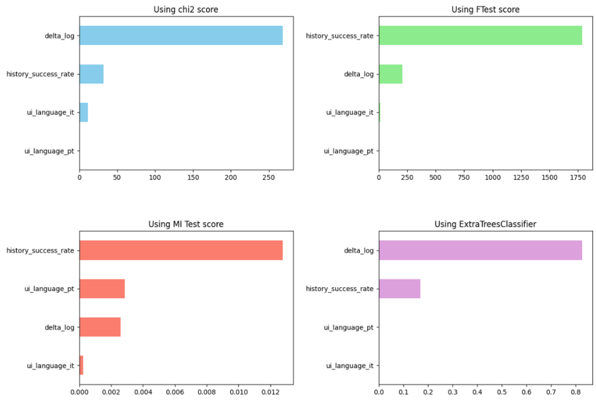


Figure 7: Comparison of Feature Importance

IV. METHODOLOGY

This project adopts a supervised learning approach as the dataset contains labelled features which can be used to train an algorithm, and a known target variable. Supervised learning is generally split into regression or classification problems. Since our target variable is categorical with only two possible outcomes, this is a classification problem. The model's accuracy is determined by how well it distinguishes between these two classes, particularly focusing on identifying the "Mistake" class amidst a majority of "Success" labels.

Logistic Regression Algorithm

Logistic Regression is a linear supervised learning algorithm used to predict the probability of a target variable belonging to a specific category. While it is a linear model, it uses a Sigmoid Function to map any real-valued number into a probability between 0 and 1.

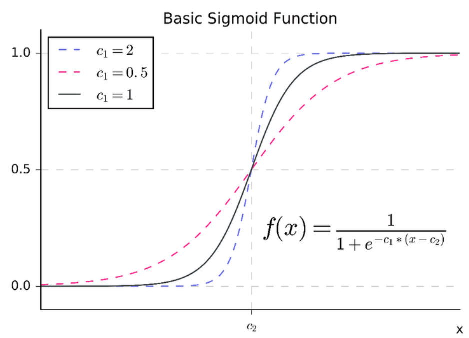


Figure 8: Sigmoid Function

For this project, Logistic Regression was selected as the initial baseline model because of its high degree of interpretability and lower training time, serving as a performance benchmark to determine if the added complexity of more advanced algorithms was necessary. As a generalized linear model it allows for direct examination of feature coefficients to determine their effect on the probability of recall, allowing us to find the optimal coefficients that maximize the likelihood of the observed data points.

Random Forest Algorithm

The Random Forest is a learning method that operates by constructing multiple Decision Trees during training. It is significantly more robust than a single decision tree because it combines the predictions of many individual trees to reach a final conclusion, with each tree trained on a random subset of the data.

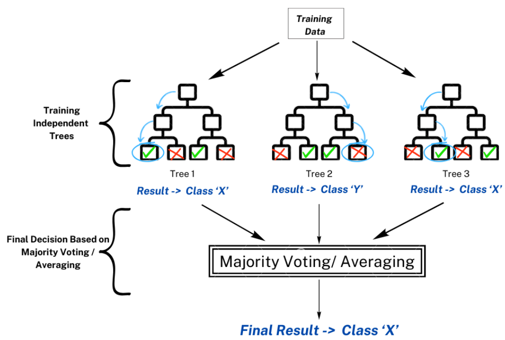


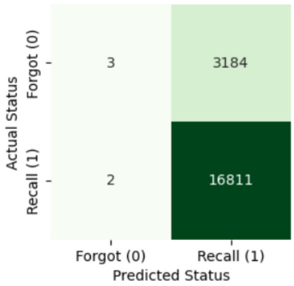
Figure 9: Random Forest

In this project, the Random Forest algorithm was chosen because it can handle the non-linear nature of the data and outliers discovered in the data exploration stage. Given the class imbalance in the Duolingo dataset, a balanced class weight of the model was utilized, forcing the algorithm to pay more attention to the minority class (Mistakes) and preventing it from simply predicting Success every time.

V. TESTING AND RESULTS

For the initial training data I split my sample dataset into an 80% training set and a 20% testing set. Given Logistic Regression's sensitivity to the magnitude of input features, feature scaling using a StandardScaler was applied to ensure that the log-transformed time gaps and historical success rates were weighted fairly. To prevent data leakage, this was fitted only on the training data and then applied to the test set, ensuring that the model's performance on the test set was a true reflection of its ability to generalize to unseen data. Cross-validation was used during the training stage to ensure the model did not overfit the specific training sample. Model performance was assessed based on accuracy, F1-score and precision.

Logistic Regression Model

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Accuracy on training data is 0.836

Test data metrics: accuracy=0.841, f1=0.913, precision=0.841
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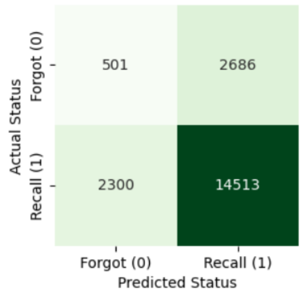
Figure 10: Logistic Regression Confusion Matrix

The Logistic Regression model serving as a linear baseline achieved a test accuracy of 0.841, with a lower training accuracy showing that it wasn't overfitting. While this accuracy appears high, the confusion matrix reveals that the model predicted Recall for nearly every instance, failing to identify almost every case where a student Forgot a word. Given the class imbalance, with the majority class making up approximately 83.7% of the data, the model's accuracy reflected the underlying class distribution rather than true prediction based on features. The decision boundary was overwhelmed by the majority class, causing the algorithm to default to a majority class prediction strategy that maximised accuracy. As such, the model is effectively useless for identifying users at risk of forgetting a word.

Random Forest Model

To address the limitations of the linear baseline, a Random Forest classifier was implemented using balanced class weights in order to penalize misclassifications of the minority class. This approach shifted the model's priority from global accuracy to recall-optimization, ensuring the system was more sensitive to the forgetting curve. Although this resulted in a lower overall accuracy of 0.751, it proved much more effective in identifying instances where a word would be forgotten, catching 505 cases. Consequently however, the model became much noisier, incorrectly predicting that 2,300 successful students would forget words. Also, the fact that the training data accuracy was significantly higher than the test data, at 0.970, suggested that the model was overfitting, memorizing training data and failing to generalize.




```
Accuracy on training data is 0.970

Test data metrics: accuracy=0.751, f1=0.853, precision=0.844
```

Figure 11: Random Forest Confusion Matrix

Hyperparameter Tuning

In order to address this overfitting, cross-validation (CV) was performed to determine the optimal maximum depth for the trees. As shown in Figure 12, this revealed a non-linear relationship between tree depth and predictive performance.

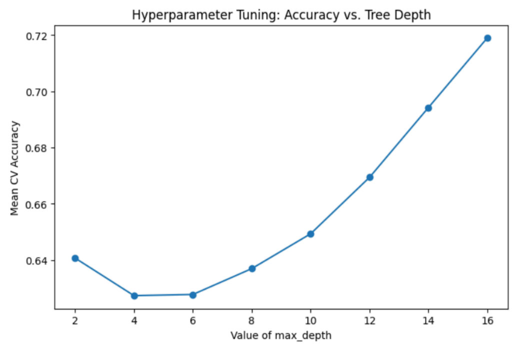


Figure 12: Hyperparameter Tuning

An initial performance decrease was observed between depths 2 and 6, suggesting a phase where the model lacked sufficient complexity to resolve the interactions between features. However, beyond depth 8, mean CV accuracy improved steadily, reaching a peak of 71.9% at depth 16 with no sign of plateauing. To maintain a balance between complexity and generalization, depth 12 was selected as the final parameter.
Applying this tuned Random Forest model resulted in a reduced overall test accuracy, falling

to 0.670. However there was a significant reduction in the gap with training accuracy of 0.716. By closing the gap between the two, the model proved more capable of generalization. This model was also more effective in identifying cases of failed word retention, with nearly 40% of all memory failures proactively identified.

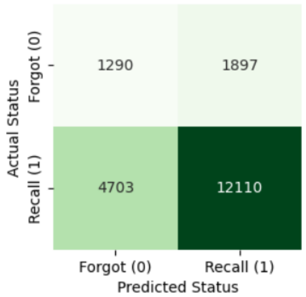


Figure 13: Tuned Random Forest

VI. CONCLUSION

This project was successful in achieving its main goal, transforming highly skewed raw interaction data into predictive variables which allowed the model to successfully predict and better understand distinct modes of learner behavior. Despite both models used having limitations, the tuned Random Forest model proved most effective at navigating class imbalance. Given the context of spaced repetition in second language learning, the cost of a false negative is significantly higher than that of a false positive, and so by optimizing for recall this model provided the most actionable means of predicting memory decay.

Processing such a large dataset, and its imbalanced nature however did somewhat limit the effectiveness of these data models, with logistic regression particularly struggling to identify cases of failed word recall. With a low average trace count per user, this study was also restricted in its ability to implement more complex user-specific modeling.

Despite this, the implications of the project outcomes are positive, with evidence that the spacing effect plays a significant role in word retention. Given a dataset focused on a smaller number of users, future studies could potentially model individual learning more accurately, and explore the impact of other demographic features on word retention, filtering data to focus on specific users with 50+ traces for instance.

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